LAB 03 Program 2 Circular Queue Using Array

Aim

To write a C program to simulate the working of a circular queue of integers using an array and perform the following operations:

1. Insert an element.
2. Delete an element.
3. Display the elements.
4. Check queue overflow and empty conditions.

```
--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter the element: 10
Element inserted successfully

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter the element: 20
Element inserted successfully

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter the element: 30
Element inserted successfully

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter the element: 40
Element inserted successfully
```

```
--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Enter the element: 50
Element inserted successfully

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 1
Queue Overflow

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 3
Queue elements are: 10 20 30 40 50

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 2
Deleted element: 10

--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 2
Deleted element: 20
```

```
--- CIRCULAR QUEUE ---
1. Insert
2. Delete
3. Display
4. Exit
Enter your choice: 4
Exiting...

Process returned 0 (0x0)   execution time : 123.889 s
Press any key to continue.
```